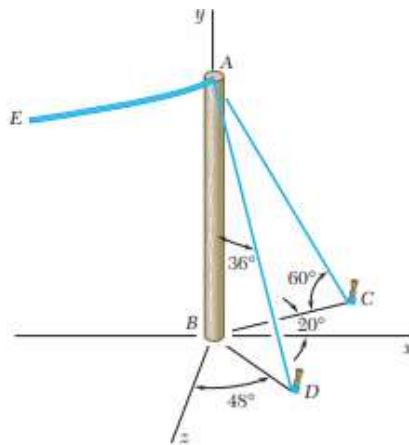
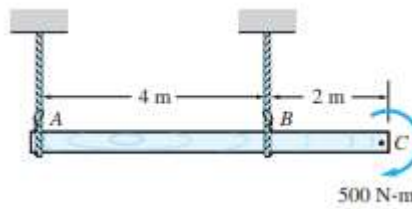


REVISION CHAPTER 2: ENGINEERING MECHANIC SET E

1. The end of the coaxial cable AE is attached to the pole AB, which is strengthened by the guy wires AC and AD. Knowing that the tension in wire AC is 120 lb, determine
 - (a) the components of the force exerted by this wire on the pole,
 - (b) the angles α , β , and γ that the force forms with the coordinate axes.

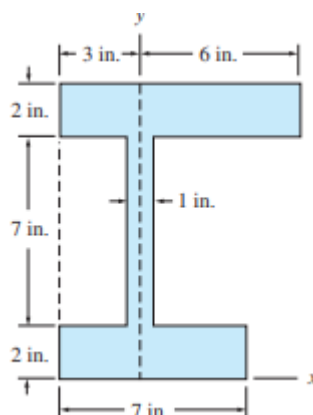


- 2.



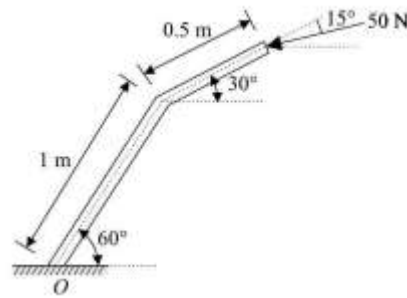
The homogeneous, 120-kg wooden beam is suspended from ropes at A and B. A power wrench applies the 500-N · m clockwise couple to tighten a bolt at C. Use the given FBD to determine the **tensions** in the ropes.

- 3.

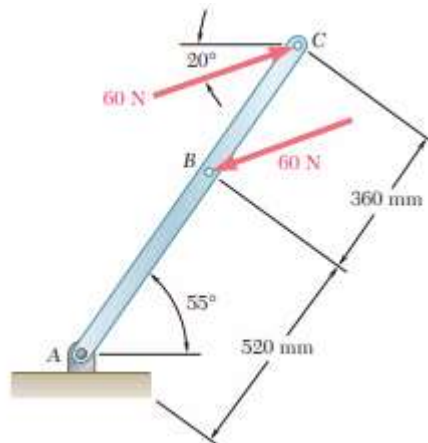


Use the method of composite areas to calculate the centroid.

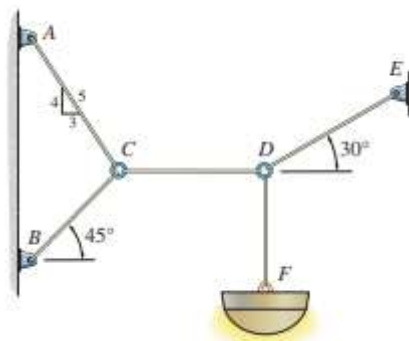
4. Find the moment at point O.



5. Two parallel 60-N forces are applied to a lever as shown. Determine the moment of the couple formed by the two forces
- (a) by resolving each force into horizontal and vertical components and adding the moments of the two resulting couples,
- (b) by using the perpendicular distance between the two forces, (c) by summing the moments of the two forces about point A .



6. a) Determine the tension developed in each cord required for equilibrium of the 20-kg lamp
- b) Determine the maximum mass of the lamp that the cord system can support so that no single cord develops a tension exceeding 400 N.



ANS

1. 56.4 lb, 103.9 lb, 20.5 lb. (b) 62.0° , 150.0° , 99.8°
2. $T = 169.3 \text{ N}$
3. $\bar{x} = 0.872 \text{ in.}$, $\bar{y} = 5.96 \text{ in}$
4. 41.83 N.m
5. 12.39 N m (b) 12.39 Nm (c) 12.39 N m
6. a) $F_{DE} = 392 \text{ N}$, $F_{CD} = 340 \text{ N}$, $F_{CB} = 275 \text{ N}$, $F_{CA} = 243 \text{ N}$, b)